



Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur

I-Mid Term Examination, March-2024

Prog./Semester:	B. Tech.- IV	Branch:	CSE/IT/AI/DS/IOT
Course Title:	DMS	Course Code:	4CS/IT/CAI/CDS/CIT2- 01
Duration:	1.5 Hours	Maximum Marks:	20
Session: First		Roll No.:	
Instructions if any:			

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	Define Partial Order Relation with an example.	2	1	1
2	Over the universe of animals, let $P(x)$: x is a whale, $Q(x)$: x is a fish, $R(x)$: x lives in water. Translate the following into English (i) $\exists x(Q(x) \wedge \sim P(x))$ (ii) $\forall x(P(x) \wedge R(x)) \Rightarrow Q(x)$	2	3	2
3	Write the contrapositive and inverse of following: If P is a square, then P is a rectangle.	2	1	2

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	In a survey of 60 people, it was found that 25 read news week, 26 read Time and 26 read magazine Fortune. Also 9 read both News week and Fortune, 11 read News Week and Time and 8 read	4	1	3

	both Time and Fortune. If 8 read none of the magazines, determine the number of people who read exactly one magazine.			
5	By means of truth table prove $(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$	4	3	3

PART- C (Descriptive/Analytical/Problem solving/Design questions)(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	Prove that dual of a Lattice is also a lattice	6	1	2
7	A relation R is defined as a R b by $a \equiv b \pmod{3}$, show that R is an equivalence relation.	6	1	2



Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.: B.Tech/IV	Course Title: DMS	Course Code:4CS/IT/CDS/CAI/CIT/2-01
Duration: 1.5 hours	Date: 28/03/24 Session: FIRST	Max Marks: 20
Submitted By: Dr. Shalini Shekhawat		

S. No.	Course Outcome	Percentage Coverage
C01	Define sets, relations, functions; Describe Hasse diagram, Lattice and apply counting operations on discrete structures in different areas of computing	57
C02	Apply mathematical techniques to prove arguments/statement/theorems; define pigeonhole principle and discuss its significance; solve problems using techniques and combinatorics	-
C03	Describe propositional Logic and test the validity of arguments, analyze finite state machine as model of physical system and language recognizers	43
C04	Classify algebraic structures, Groups, Rings and Fields; solve problems using algebraic structures	-
C05	Define graphs; classify graphs by their characteristics, apply graph theory to solve problems in different areas of computer science	-

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	14
2	BL2- Understand	57
3	BL3- Apply	29
4	BL4- Analyze	--
5	BL5- Evaluate	--
6	BL6- Create	--

Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.: B.Tech/IV	Course Title: DMS	Course Code: 4CS/IT/CDS/CAI/CIT/2-01
Duration: 1.5 hours	Date: 28/03/24 Session: FIRST	Max Marks: 20
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-- Question Paper Solution --

Ans. 1 A relation R defined on a set A is said to be partial order relation in A if R is reflexive, R is anti-symmetric, R is transitive.

Ex: If R be a relation in set N of natural numbers defined by " x divides y ". Then N is a partial order Set.

Ans. 2

(i) There exists a fish that is not whale.

(ii) Every whale that lives in the water, is a fish.

Ans. 3.

(i) If P is not a rectangle, then P is not a square.

(ii) If P is not a square, then P is not a rectangle.

Ans 4.

Let $V = 8$

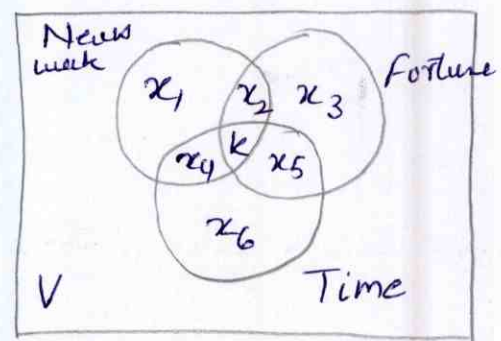
$$(x_1 + x_3 + x_6) + (x_2 + x_4 + x_5) + k = 60 - 8 = 52 \quad \text{--- (1)}$$

$$x_1 + x_2 + x_4 + k = 25 \quad \text{--- (2)}$$

$$x_4 + (x_5 + x_6) + k = 26 \quad \text{--- (3)}$$

$$x_2 + (x_3 + x_5) + k = 26 \quad \text{--- (4)}$$

$$x_2 + k = 9 \quad \text{--- (5)}$$





Solution of Question Paper
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$$x_4 + k = 11 \quad \text{--- (6)}$$

$$x_5 + k = 8 \quad \text{--- (7)}$$

$$n(U) = 60$$

On adding (5), (6) & (7)

$$x_2 + x_4 + x_5 + 3k = 28 \quad \text{--- (8)}$$

Adding (2), (3) and (4)

$$(x_1 + x_3 + x_6) + 2(x_2 + x_4 + x_5) + 3k = 77$$

$$\text{Let } x_1 + x_3 + x_6 = p \quad x_2 + x_4 + x_5 = q$$

$$p + q + k = 52$$

$$q + 3k = 28$$

On solving we get

$$k = 3, p = 30, q = 19$$

Those who read only one magazine = $x_1 + x_3 + x_6 = p = 30$.

Ans 5.

P	Q	R	$P \cap Q$	$P \cup R$	$P \cap Q \cup R$	$Q \cup R$	$(P \cap Q) \cup R$
T	T	T	T	T	T	T	T
T	F	F	F	T	F	F	F
F	T	F	F	F	T	F	F
F	F	T	F	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	F	T	T	T	T
F	T	T	F	T	T	T	T
F	F	F	F	F	F	F	F



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As we can see the last two columns have same values of T & F so

$$(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$$

Ans. 6 Let (L, R) be a lattice, where R is a ~~partial~~ order relation
i.e. (L, R) is a Poset.

If R^{-1} is inverse of R then R^{-1} is also a partial order relation.

$\Rightarrow (L, R^{-1})$ is also a Poset.

Now we will prove that for any $a, b \in L$, $a \vee b$ and $a \wedge b$ are the Glb and lub resp. of the subset $\{a, b\}$ with regard to R^{-1} .

Since $a, b \in L$

with regard to R , $a \vee b = \text{lub of } \{a, b\}$

$$a R (a \vee b) \text{ and } b R (a \vee b)$$

$$\Rightarrow (a \vee b) R^{-1} a \text{ and } (a \vee b) R^{-1} b$$

$$\Rightarrow (a \vee b) \text{ is lb of } a \text{ \& } b \text{ with regard to } R^{-1}$$

Now let c is another lb of a & b with regard to R^{-1} then

$$c R^{-1} a \text{ and } c R^{-1} b \Rightarrow a R c \text{ and } b R c$$

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$\Rightarrow c$ is ub of a and b with regard to R

$$\Rightarrow (a \vee b) R c$$

$$\Rightarrow c \in R^{-1}(a \vee b)$$

$\Rightarrow a \vee b$ is the glb of a and b with regard to R .

So in (L, R^{-1}) $a \vee b = \text{glb} \{a, b\}$

Similarly we can show in (L, R^{-1})

$$a \wedge b = \text{lub} \{a, b\}$$

Hence (L, R^{-1}) is a lattice.

Ans. 7. Given that $x R y \Leftrightarrow x - y$ is divisible by 3 $\forall x, y \in \mathbb{N}$

1. Reflexive:- Since $x - x = 0 \forall x \in \mathbb{N}$

and 0 is divisible by 3

$$x R x \forall x \in \mathbb{N}$$

$\Rightarrow R$ is reflexive.

2. Symmetric:- Let $x R y \Rightarrow x - y$ is divisible by 3

$\Rightarrow -(x - y)$ is also divisible by 3.

$\Rightarrow y - x$ is divisible by 3

$\Rightarrow y R x$
 $\Rightarrow R$ is symmetric



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Transitive:- Let xRy and yRz then

$$x - y = 3n_1 \quad \text{for } n_1, n_2 \in \mathbb{Z}$$

$$y - z = 3n_2$$

on adding these equations

$$x - z = 3(n_1 + n_2)$$

$$x - z = 3n, \quad n \in \mathbb{Z}$$

$\Rightarrow x - z$ is divisible by 3

$$\Rightarrow xRz$$

$\Rightarrow R$ is transitive

Hence R is an equivalence relation.



Swami Keshvanand Institute of Technology, Management
& Gramothan, Jaipur

I-Mid Term Examination, March-2024

Prog./Semester:	B. Tech.-IV	Branch:	IT/CE/EC/EE/ME
Course Title:	Technical Communication	Course Code:	4IT/CE/EC/ME 1-02
Duration:	1.5 Hours	Maximum Marks:	20
Session : First/Second/Third	Roll No.:		
Instructions if any:			

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	What is conciseness in technical communication?	2	1	1
2	Describe the importance of technical communication.	2	1	2
3	Explain the importance of 'organisational competence' in technical communication.	2	1	2

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Demonstrate the importance of Cornell method of note-taking with the help of an example.	4	2	3
5	What strategy one should follow to select an article appropriate for the given research purpose.	4	2	4

PART- C (Descriptive/Analytical/Problem solving/Design questions) (Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	How does mind mapping differ from traditional linear note-making method?	6	2	4
7	Define electronic communication. In what ways it is more advantageous than oral and written forms.	6	2	4



Swami Keshvanand Institute of Technology, Management & Gramothan,
Ramnagar, Jagatpura, Jaipur-302017

Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.: B.Tech IV Sem	Course Title: Technical Communication	Course Code:4IT/CE/EC/EE/ME 1-02
Duration: 1.5 hours	Date: 22/3/24 Session : SECOND	Max Marks: 20
Submitted By: Dr Anupriya Singh		

S. No.	Course Outcome	Percentage Coverage
C01	Recall and identify four basic LSRW skills for learning technical communication.	23.07
C02	Read, understand and summarize the technical texts and documents.	76.92
C03	Apply technical style in writing and note-making.	
C04	Compose official notes, letters, emails, resume, job application and MoM with the usage of advanced grammar.	
C05	Design, analyze, and evaluate technical reports, articles and proposal in proper formats.	

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	7.69
2	BL2- Understand	15.38
3	BL3- Apply	15.38
4	BL4- Analyze	61.53
5	BL5- Evaluate	nil
6	BL6- Create	nil

-- Question Paper Solution --



Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.: B.Tech IV Sem	Course Title: Technical Communication	Course Code: 4IT/CE/EC/EE/ME 1-02
Duration: 1.5 hours	Date: 22/3/24 Session : SECOND	Max Marks: 20
Submitted By: Dr Anupriya Singh		

Q1. What is conciseness in technical communication?

Ans. Conciseness in technical communication refers to the practice of conveying information in a clear and efficient manner, using as few words as necessary to communicate the intended message. This involves avoiding unnecessary details, repetition, and verbosity, while ensuring that the content remains complete and understandable to the intended audience. Conciseness helps to maintain the reader's attention, improves understanding, and enhances the overall effectiveness of the communication.

Q2. Describe the importance of technical communication

Ans. Technical communication is crucial as it facilitates the transfer of complex information in a clear, concise, and understandable manner. It plays a vital role in various fields, including technology, science, engineering, and business, ensuring that ideas, instructions, and data are effectively communicated to a diverse audience. Effective technical communication improves productivity, reduces errors, enhances decision-making, and fosters innovation. It also helps to build credibility, trust, and collaboration among individuals and teams.

Q3. Explain the importance of 'organisational competence' in technical communication.

Ans. Organizational competence in technical communication refers to the ability of an organization to effectively communicate technical information both internally and externally. It encompasses the skills, processes, and strategies employed by the organization to ensure that technical communication meets its objectives and serves its stakeholders.

Organizational competence is crucial for several reasons:

1. **Efficient Operations:** Clear and effective technical communication ensures that employees understand their roles and responsibilities, leading to more efficient operations within the organization.
2. **Product Development:** Technical communication plays a crucial role in product development, helping teams collaborate effectively and ensuring that products meet user needs and specifications.
3. **Customer Satisfaction:** Clear technical communication helps customers understand products or services, leading to higher satisfaction and fewer support requests.
4. **Compliance:** In regulated industries, technical communication is essential for ensuring compliance with laws, regulations, and standards.
5. **Risk Management:** Effective technical communication can help identify and mitigate risks associated with products, processes, or services.

Overall, organizational competence in technical communication is essential for ensuring that information is conveyed accurately, clearly, and efficiently, benefiting both the organization and its stakeholders.



Solution of Question Paper
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Prog./Sem.: B.Tech IV Sem	Course Title: Technical Communication	Course Code:4IT/CE/EC/EE/ME 1-02
Duration: 1.5 hours	Date: 22/3/24 Session : SECOND	Max Marks: 20
Submitted By: Dr Anupriya Singh		

Q4. Demonstrate the importance of Cornell method of note-taking with the help of an example.

Ans. The Cornell method of note-taking is a systematic approach that divides notes into three sections: a narrow column on the left for key points or cues, a wider column on the right for detailed notes, and a summary section at the bottom. This method is effective for organizing information, aiding in comprehension, and facilitating review.

For example, let's consider a student taking notes during a lecture on the topic of "The Water Cycle":

1. **Cues/Key Points (Left Column):**

- Precipitation
- Evaporation
- Condensation
- Runoff

2. **Notes (Right Column):**

- Precipitation: Water falls to the ground as rain, snow, sleet, or hail.
- Evaporation: Sun heats water on the Earth's surface, turning it into water vapor.
- Condensation: Water vapor cools and changes back into liquid, forming clouds.
- Runoff: Excess water flows over the Earth's surface into lakes, rivers, and oceans.

3. **Summary (Bottom Section):**

- The water cycle is the continuous process of water circulating between the atmosphere, land, and bodies of water.
- It is driven by solar energy and is crucial for maintaining Earth's ecosystems and supporting life.

In this example, the Cornell method helps the student organize key points, detailed information, and a summary, making it easier to review and understand the water cycle. The cues/key points section serves as a quick reference, while the notes section provides detailed explanations. The summary section helps in synthesizing the information and reinforcing key concepts.

Q5. What strategy one should follow to select an article appropriate for the given research purpose.

1. **Define the Research Purpose:** Clearly define your research question or objective. This will help you identify the specific criteria the article should meet.
2. **Identify Relevant Keywords:** Use relevant keywords related to your research topic to search for articles in databases, libraries, or online sources.
3. **Use Filters:** Use filters such as publication date, type of publication (e.g., journal article, conference paper), and study design (e.g., experimental, qualitative) to narrow down your search results.
4. **Evaluate the Source:** Evaluate the credibility and reliability of the source. Consider factors such as the author's credentials, the reputation of the publication or journal, and whether the article has been peer-reviewed.
5. **Read the Abstract:** Read the abstract of the article to get an overview of the study, its methodology, and its findings. This will help you determine if the article is relevant to your research purpose.
6. **Scan the Introduction and Conclusion:** Scan the introduction and conclusion of the article to get a sense of the author's argument, the scope of the study, and its relevance to your research.
7. **Check References:** Check the references cited in the article to find other relevant sources that you can use for your research.



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8. **Consider the Currency:** Consider how recent the article is. Depending on your research topic, you may need the most up-to-date information or historical perspectives.
9. **Seek Diverse Perspectives:** If your research benefits from different viewpoints, consider including articles that present diverse perspectives on the topic.
10. **Consult with Peers or Mentors:** Discuss your article selection with peers, mentors, or librarians to get feedback and ensure that your choices align with your research purpose.

Q6. How does mind mapping differ from traditional linear note-making method?

Ans. Mind mapping differs from traditional linear note-making methods in several ways:

1. **Structure:** Mind maps use a radial, non-linear structure, with the main topic at the center and subtopics branching out from it. This allows for a more visually organized representation of information compared to the linear format of traditional notes.
2. **Creativity:** Mind mapping encourages creativity and association of ideas. Users can add colors, images, and symbols to represent concepts, making it a more engaging and stimulating way to take notes compared to the more text-heavy traditional method.
3. **Flexibility:** Mind maps are highly flexible and can easily accommodate new ideas or connections. It is easy to add or rearrange branches and subtopics, allowing for a dynamic representation of information.
4. **Visualization:** Mind maps provide a visual overview of the topic, making it easier to see relationships between different ideas and concepts. This visual representation can aid in understanding complex information and in generating new ideas.
5. **Memory:** The visual and spatial layout of mind maps can aid in memory retention. The use of images and colors can help make information more memorable compared to the linear format of traditional notes.
6. **Holistic Thinking:** Mind mapping encourages holistic thinking by showing the interconnectedness of ideas. This can help users see the bigger picture and understand how different concepts relate to each other.

Overall, mind mapping offers a more creative, flexible, and visually engaging alternative to traditional linear note-making methods, making it a valuable tool for brainstorming, organizing information, and enhancing understanding and memory.



Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.: B.Tech IV Sem	Course Title: Technical Communication	Course Code:4IT/CE/EC/EE/ME 1-02
Duration: 1.5 hours	Date: 22/3/24 Session : SECOND	Max Marks: 20
Submitted By: Dr Anupriya Singh		

Q7. Define electronic communication. In what ways it is more advantageous than oral and written forms.

Ans. Electronic communication refers to the exchange of information, ideas, or messages using electronic devices and technologies. This includes email, instant messaging, social media platforms, video conferencing, and other digital communication tools.

Electronic communication offers several advantages over oral and written forms:

1. **Speed:** Electronic communication allows for almost instantaneous transmission of messages, enabling real-time communication over long distances.
2. **Convenience:** It allows people to communicate asynchronously, meaning they can send and receive messages at their convenience, without needing to be available at the same time.
3. **Cost-Effective:** Compared to traditional methods such as postal mail or phone calls, electronic communication is often more cost-effective, especially for long-distance or international communication.
4. **Accessibility:** Electronic communication can be accessed from various devices, including smartphones, tablets, and computers, making it easier for people to stay connected regardless of their location.
5. **Storage and Retrieval:** Messages and information exchanged electronically can be easily stored, organized, and retrieved, making it easier to keep track of conversations and reference past communication.
6. **Multimedia Support:** Electronic communication allows for the exchange of not just text but also multimedia content such as images, videos, and documents, enhancing the richness of communication.
7. **Global Reach:** It enables communication with people around the world, breaking down geographical barriers and facilitating global collaboration and networking.
8. **Environmental Impact:** Compared to traditional paper-based communication, electronic communication has a lower environmental impact, as it reduces the need for paper and transportation.

Despite these advantages, electronic communication also has some limitations, such as the potential for misinterpretation of messages due to the lack of non-verbal cues, security and privacy concerns, and the risk of information overload.



**Swami Keshvanand Institute of
Technology, Management & Gramothan,
Jaipur**

I -Mid-Term Examination, March 2024

Prog./Semester :	B. Tech.-IV	Branch:	IT
Course Title:	POC	Course Code:	4IT3-04
Duration:	1.5 Hours	Maximum Marks:	20
Session: First		Roll No.:	
Instructions if any:			

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	Why do we need modulation in communication?	2	1	2
2	Draw the waveforms of AM and FM with fixed modulating signal and carrier signal.	2	2	2
3	State the sampling theorem and explain the reason for sampling.	2	1	2

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	An FM wave is given by. $s(t)=12 \sin (5 \times 10^8 t+7 \sin 1350 t)$ Determine the following:	1	2	3
	i. The carrier frequency and Modulating frequency			
	ii. Maximum frequency deviation	1		
	iii. Modulation index	1		
	iv. FM Bandwidth	1		

5	Calculate the Nyquist frequency and Nyquist interval when a continuous-time signal is given by: $X(t) = \cos(5000\pi t) \cdot \cos(1000\pi t)$	4	1	3
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PART- C (Descriptive/Analytical/Problem solving/Design questions)
(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	Justify that the balance slope detector is more efficient compared to the slope detector for FM detection with the help of a suitable diagram.	6	2	3
7	i. Indicate how the carrier is suppressed in DSB-SC and discuss the benefits and drawbacks of this modulation technique.	3	2	3
	ii. Draw the waveform of the AM signal for the value of modulation index is less than one, equal to one, and greater than one ($m_a < 1$, $m_a = 1$, $m_a > 1$).	3		



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Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.:	Course Title: POC	Course Code: 4IT3-04
Duration: 1.5 hours	Date: 30 March 2024. Session: FIRST	Max Marks: 20
Submitted By: Dr. J.P Vijay, Dr. Ankit Agarwal, Gloria Joseph		

S. No.	Course Outcome	Percentage Coverage
CO1	Understand the fundamental principles, concepts, and important components of analog and digital communication systems in various technological contexts.	30.7
CO2	Outline the various analog and digital modulation techniques for communication systems.	69.3
CO3	Demonstrate the ability to select appropriate analog-to-digital conversion methods for specific scenarios.	
CO4	Apply knowledge to implement effective analog and digital communication systems.	
CO5	Explain the fundamental principles, mechanisms, and applications of spread-spectrum modulation.	

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	-
2	BL2- Understand	23%
3	BL3- Apply	77%
4	BL4- Analyze	
5	BL5- Evaluate	
6	BL6- Create	

– Question Paper Solution –



Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.:	Course Title: POC	Course Code: 4IT3-04
Duration: 1.5 hours	Date: 30 March 2024. Session: FIRST	Max Marks: 20
Submitted By: Dr. J.P Vijay, Dr. Ankit Agarwal, Gloria Joseph		

Q.1: Why do we need modulation in communication?

Ans: Need of Modulation:

- Reduction in the height of antenna
- Avoids mixing of signals.
- Increases the range of communication.
- Multiplexing is possible.
- Improves quality of reception

Q.2: Draw the waveforms of AM and FM with fixed modulating signal and carrier signal.

Ans: waveforms of AM and FM

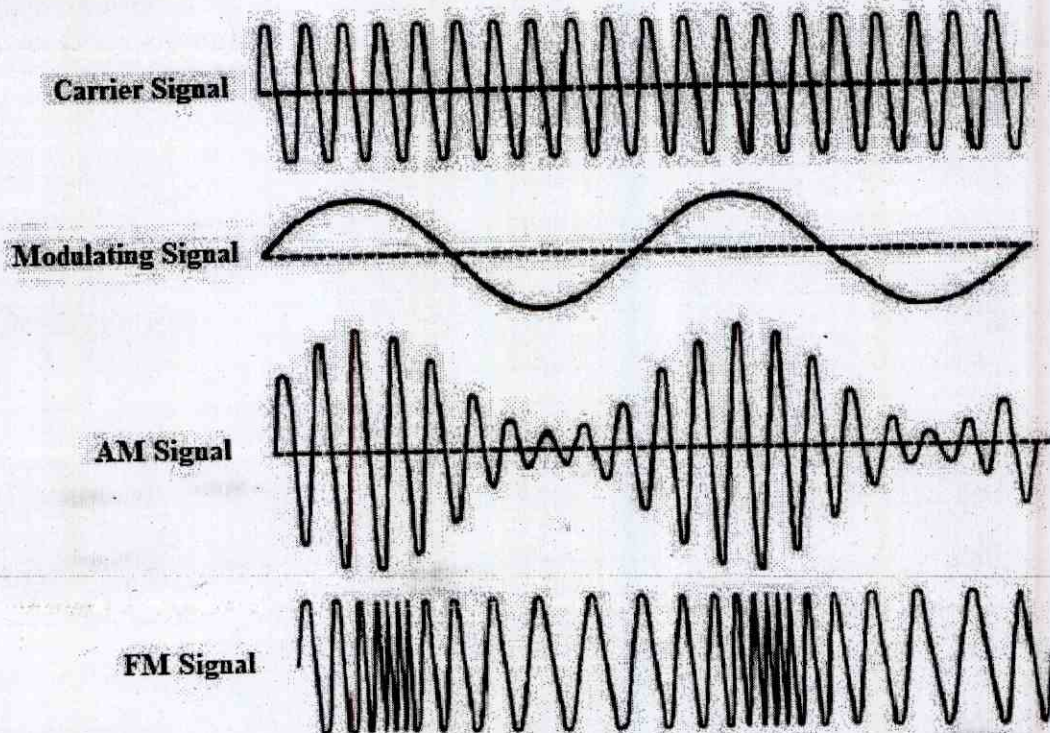


Figure 1. Waveforms of AM and FM

Q.3. State the sampling theorem and explain the reason of sampling.

Ans: The sampling theorem, also known as the Nyquist-Shannon sampling theorem, states that:



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Submitted By: Dr. J.P Vijay, Dr. Ankit Agarwal, Gloria Joseph		

"A continuous-time signal can be accurately represented by its samples and recovered back, if the sampling rate (F_s) is greater than twice the maximum frequency component ($f_{\max}$) present in the signal, which is also known as the Nyquist rate."

Mathematically, the sampling theorem can be expressed as follows: $F_s \geq 2f_{\max}$

The reason for sampling in communication and signal processing is to convert continuous analog signals into discrete time signals. There are several key reasons for doing this:

- **Efficient Transmission and Storage:** Digital signals are easier to transmit and store than analog signals. Sampling converts the analog signal into a sequence of digital samples, making it suitable for transmission over digital communication channels and storage in digital memory devices.
- **Noise Reduction and Signal Processing:** Analog signals are susceptible to noise and distortion during transmission. Sampling enables the application of various signal processing techniques, such as filtering, modulation, and error correction coding, to improve the signal quality and reduce the effects of noise.
- **Interoperability and Compatibility:** Sampling facilitates interoperability and compatibility between different communication systems and devices. Once the analog signal is converted into digital form, it can be transmitted, processed, and interpreted using standardized digital communication protocols and formats. This ensures seamless communication between diverse systems and enables integration with digital technologies.
- **Accuracy and Precision:** Sampling allows for the accurate representation of analog signals with minimal loss of information. By appropriately choosing the sampling rate according to the Nyquist-Shannon sampling theorem, it is possible to faithfully capture the essential characteristics of the original analog signal. This ensures high-quality signal reproduction and reliable communication.

Sampling is fundamental in communication systems as it enables the efficient and reliable transmission of information in various forms while overcoming challenges such as noise, bandwidth limitations, and signal processing requirements.

Q.4. An FM wave is given by $s(t)=12 \sin (5 \times 10^8 t+7 \sin 1350 t)$

Determine

- The carrier frequency and Modulating frequency**
- Maximum frequency deviation**



Solution of Question Paper
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Prog./Sem.:	Course Title: POC	Course Code: 4IT3-04
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Submitted By: Dr. J.P Vijay, Dr. Ankit Agarwal, Gloria Joseph		

iii. Modulation index

iv. FM Bandwidth

Ans: Given (by comparison with standard equation of FM): $A = 12V$, $\omega_c = 5 \times 10^8$ radian/sec $m_f = 7$
 $\omega_m = 1350$ radian/sec

Find

- The carrier frequency $f_c = \omega_c / 2\pi = (5 \times 10^8) / (2 \times 3.14) = 0.80 \times 10^8$ Hz
Modulating frequency $f_m = \omega_m / 2\pi = (1350) / (2 \times 3.14) = 215$ Hz
- Modulation index $m_f = 7$
- Maximum frequency deviation $\Delta f = m_f \cdot f_m = 7 \times 215 = 1.50$ KHz:
- FM Bandwidth $= 2 (\Delta f + f_m) = 2 (1.50 + 0.215) = 3.43$ KHz

Q.5. Calculate the Nyquist frequency and Nyquist interval when a continuous-time signal is given by:

$$X(t) = \cos(5000 \pi t) \cdot \cos(1000 \pi t)$$

Ans: Given continuous-time signal $X(t) = \cos(5000 \pi t) \cdot \cos(1000 \pi t)$

Since we know that $\cos A \cdot \cos B = \cos(A+B) + \cos(A-B)$

$$X(t) = \cos(5000 \pi t) \cdot \cos(1000 \pi t) = \cos(6000 \pi t) + \cos(4000 \pi t)$$

Since signal can be define as $x(t) = V_m \cdot \cos(2\pi f t)$

By comparing

$$f_1 = 3000 \text{ Hz}$$

$$f_2 = 2000 \text{ Hz}$$

To find Nyquist frequency $F_s = 2F_{\max}$

Since for given signal maximum frequency $F_{\max} = 3000$ Hz

$$\text{Nyquist frequency } F_s = 2F_{\max} = 2 \times 3000 = 6000 \text{ Hz} \quad \text{Ans}$$

$$\text{Nyquist Interval } T_s = 1 / F_s = 1 / 2F_{\max} = 1/6000 = 0.166 \text{ ms.} \quad \text{Ans}$$

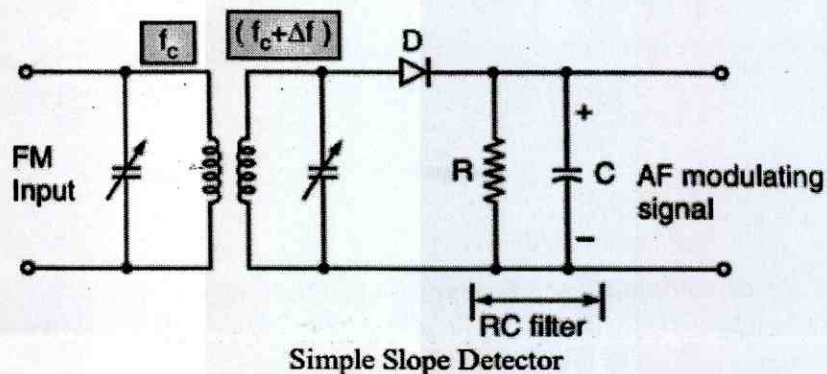


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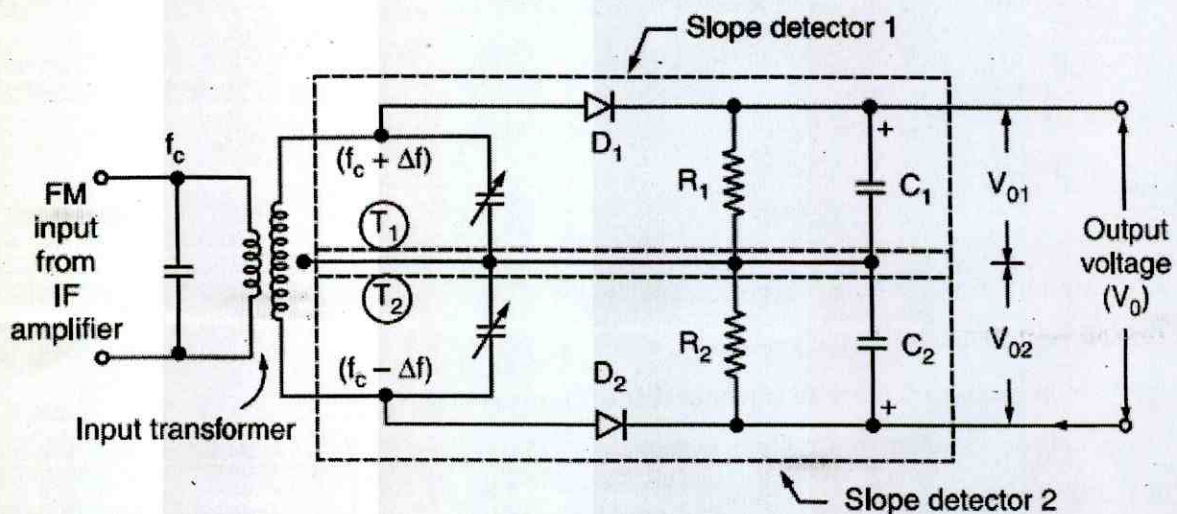
Prog./Sem.:	Course Title: POC	Course Code: 4IT3-04
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Submitted By: Dr. J.P Vijay, Dr. Ankit Agarwal, Gloria Joseph		

Q.6: Justify that the balance slope detector is more efficient compared to the slope detector for FM detection with the help of a suitable diagram.

Ans. Slope Detector: This is a circuit or algorithm designed to detect changes or slopes in a signal. It's commonly used in applications where detecting transitions or changes in a signal is important.



Balance Slope Detector: This is a variation of the slope detector that operates with balanced inputs. Balanced inputs mean that the detector is designed to handle signals that have equal but opposite polarities, typically used in differential signaling.



Balance slope detectors might be more efficient due to the following reasons.

- This circuit is more efficient than a simple slope detector because the slope detector is tuned at a frequency of $f_c + \Delta f$ whereas the balanced slope detector is tuned for the $f_c + \Delta f$ and $f_c - \Delta f$ frequency.



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- **Common-Mode Rejection:** Balanced inputs allow for better rejection of common-mode noise. Common-mode noise refers to interference that affects both signal lines equally. By using balanced inputs, the detector can effectively cancel out such noise, leading to more accurate detection of signal slopes.
- **Improved Signal Integrity:** In many practical scenarios, signals are transmitted differentially for improved noise immunity and signal integrity. A balanced slope detector is inherently suitable for such scenarios as it can directly process these differential signals without needing additional conversion or processing stages.
- **Higher Sensitivity:** With balanced inputs, the detector can have higher sensitivity to small changes in the signal. This is because it can effectively amplify the differential component of the signal while rejecting common-mode noise, leading to improved detection of subtle changes in the signal slope.
- **Reduced False Positives/Negatives:** The improved noise rejection and sensitivity of a balance slope detector generally result in fewer false positives or false negatives in slope detection tasks. This is crucial in applications where accurate detection of signal transitions is paramount.
- **Lower Power Consumption:** In many cases, balance slope detectors can achieve their objectives with lower power consumption compared to non-balanced counterparts. This efficiency is particularly important in battery-operated or low-power devices.

In summary, the balance slope detector offers advantages in terms of noise rejection, sensitivity, false detection rates, power consumption, and compatibility with modern communication interfaces, making it a more efficient choice compared to a regular slope detector in many practical scenarios.

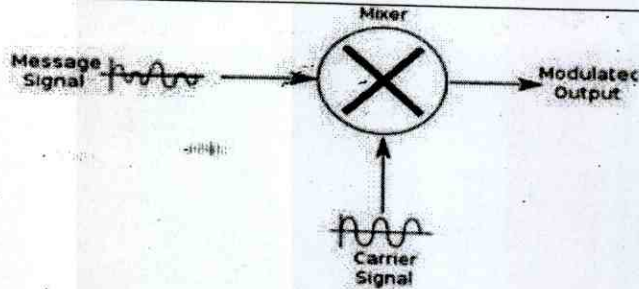
Q.7: Indicate how the carrier is suppressed in DSB-SC, and discuss the benefits and drawbacks of this modulation technique. Draw the waveform of AM signal for the value of modulation index is less than one, equal to one and greater than one ($m_a < 1$, $m_a = 1$, $m_a > 1$).

Sol: DSB-SC is the modulation technique by suppresses the carrier signal for maximum power utilization. For the transmission of carrier signal, more power is required but this power is not used for the transmission of signal. So the power may be suppressed for minimize the wastage of power and this signal is generated by a mixer. This consists of a message signal multiplied by a carrier signal. The mixing of the message signal and carrier signal is shown in the figure below.



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For the suppression of carrier signal and generation of DSB-SC modulation signal, two modulation techniques is used.

1. Balanced Modulator
2. Ring Modulator

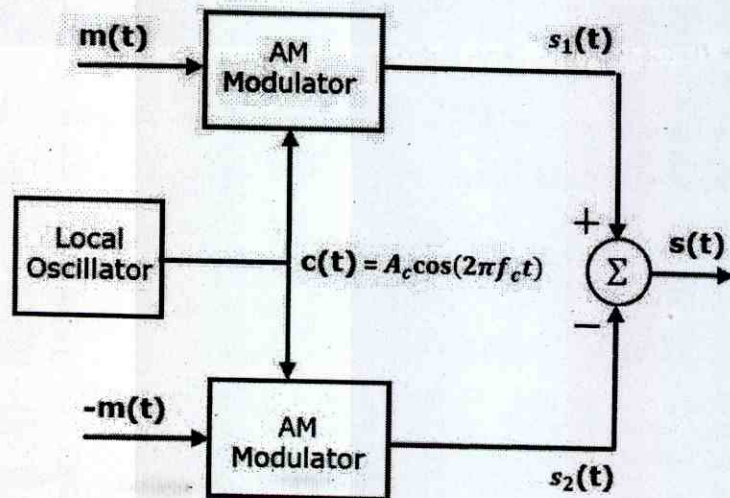


Figure: Balanced Modulator



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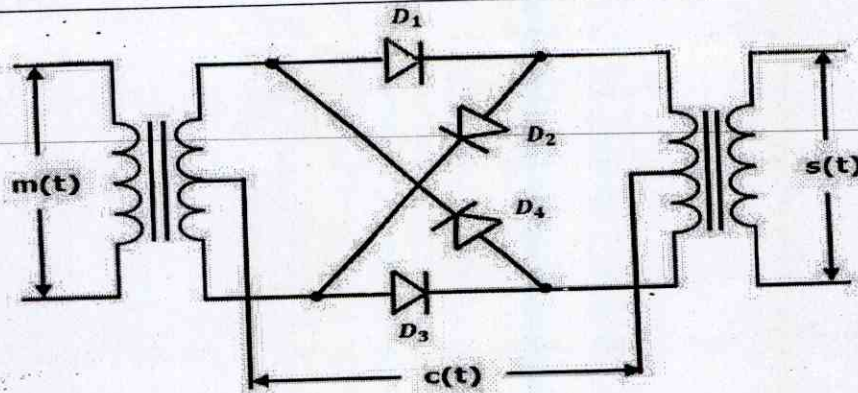


Figure: Ring Modulator

Benefits of DSB-SC Modulation

- DSB-SC provides 50% modulation efficiency due to the absence of a carrier.
- It consumes less power again due to the absence of a carrier in DSB-SC.
- It provides a large bandwidth due to the presence of two sidebands. The output frequency is twice the frequency of the modulated signal.
- Low-cost design.

Disadvantages of DSBSC

The disadvantages of DSB are as follows:

- The efficiency of DSB with the carrier is low due to the presence of the carrier.
- The power wastage is high in DSB compared to other types of Amplitude Modulation because of the two sidebands that are not effectively utilized.
- It becomes difficult to recover the information at the receiving end. The recovery of the signal depends on the carrier. If the carrier is absent, it needs to be regenerated. Such process of recovery at the receiver becomes complex.



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Modulation Index / Depth of modulation:

The amount by which the amplitude of the carrier wave increases and decreases depends on the amplitude of the information signal and is called the “depth of modulation”. The depth of modulation can be quoted as a fraction or as a percentage.

Formula Used for modulation index:

$$\% \text{ Modulation} = \frac{V_m}{V_c} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} * 100$$

V_m is the maximum amplitude of the modulating signal; V_c is the maximum amplitude of carrier signal.

Waveform of AM: For a perfect modulation, the value of modulation index should be 1, which means the modulation depth should be 100%.

If this value is less than 1, i.e., the modulation index is 0.5, then it is called as Under-modulation. Such a wave is called as an under-modulated wave.

If the value of the modulation index is greater than 1, i.e., 1.5 or so, then the wave will be an over-modulated wave.



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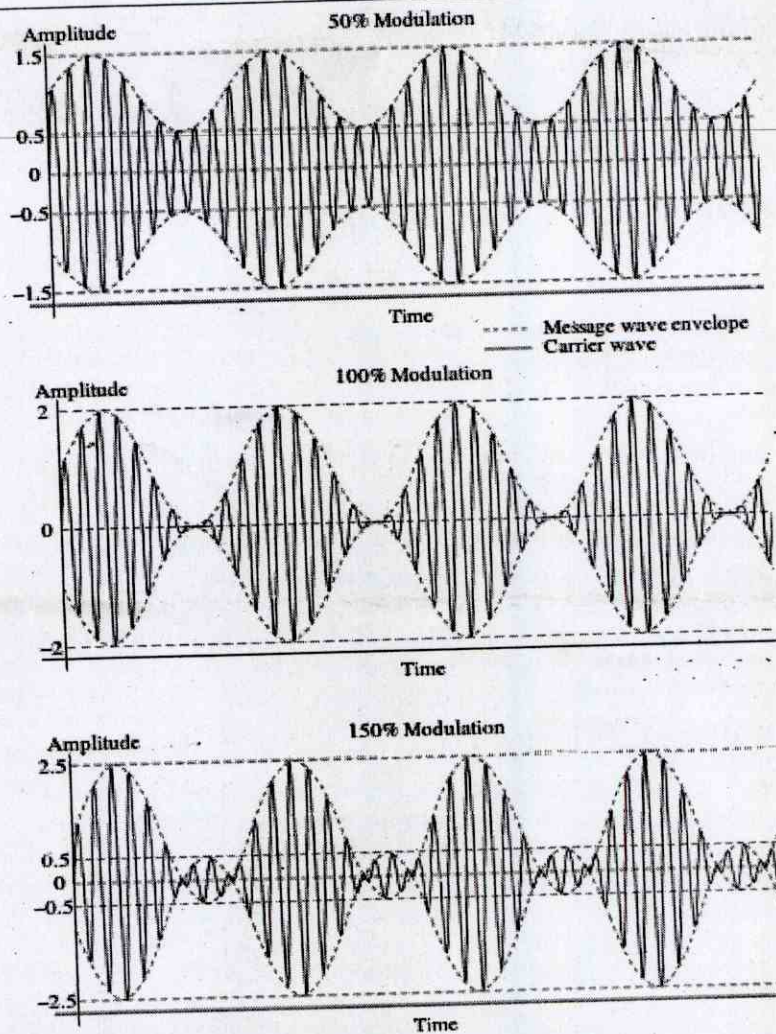


Figure: AM waveform for different modulation index



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Prog./Semester:	B. Tech.-IV	Branch:	CSE,IT,DS,AI,IOT
Course Title:	Database Management System	Course Code:	4CS4-05, 4IT4-05, 4CDS4-05, 4CAI4-05, 4CIT4-05
Duration:	1.5 Hours	Maximum Marks:	20
Session : First	Roll No.:		
Instructions if any:			

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO (s)	Bloom's Level
1	Define RDBMS.	2	CO1	L1
2	Explain types of attributes.	2	CO1	L2
3	Explain the role of join operation?	2	CO2	L2

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Illustrate all basic relational algebra operations. Consider following relations and analyze relational algebra Queries: Suppliers (sID, sName, address) , Parts (pID, pName, colour), Catalog (sID, pID, price) 1) Find all prices for parts that are red or green. (A part may have different prices from different manufacturers.) 2) Find the names of all suppliers who supply a part that is red or green.	4	CO2	L4
5	Demonstrate advantageous of DBMS over File System.	4	CO1	L3

PART- C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	Show a neat Diagram of DBMS System Structure also outlines all the parts of it.	6	CO1	L3
7	Model the E-R diagram for library management system with complete labelling. Convert the drawn E-R diagram into relational model. For each relation in relational model clearly specify primary key and foreign key.	6	CO1	L3



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**Solution of Question Paper
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Prog./Sem.: CSE, AI, DS, IoT, IT/ IV	Course Title: Database Management System	Course Code: 4CS4-05, 4IT4-05, 4CDS4-05, 4CAI4 -05, 4CIT4-05
Duration: 1.5 hours	Date: 30/3/24 Session : FIRST/SECOND/THIRD	Max Marks: 20
Submitted By: Dr. Kajal, Ms. Anjali, Mr. Harpreet, Mr. Kailash, Ms. Mamta, Mr. Vikram, Ms Suchita,		

S. No.	Course Outcome	Percentage Coverage
C01	<i>Explain</i> the basic concepts of RDBMS, relational data model, entity-relationship model and relational database design	76.92%(20)
C02	<i>Apply</i> the relational database theory to write basic and advanced SQL queries and relational algebra and calculus expressions for the queries.	30% (6)
C03	<i>Classify</i> the types of normalization and functional dependency in database design.	
C04	<i>Apply</i> the concept of transactions and concurrency control protocols along with the various recovery methods	

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	7.69%(2)
2	BL2- Understand	15.38%(4)
3	BL3- Apply	61.53%(16)
4	BL4- Analyze	15.38%(4)
5	BL5- Evaluate	
6	BL6- Create	

Q.1 Define RDBMS.

[2 marks]

Ans: - RDBMS stands for Relational Database Management System. It's a type of database management system that organizes data into tables of rows and columns, with each row uniquely identified by a key. It enables users to establish relationships between different tables and perform operations like querying, updating, and managing data efficiently.

Q.2 Explain types of attributes?

[2 marks]

Ans:- Types of attributes in a database include:

- Simple attribute: Atomic values that cannot be divided further, such as an integer or a string.
- Composite attribute: Attributes that can be divided into smaller sub-parts, each representing some simpler attribute with its own meaning.
- Single-valued attribute: Attributes that hold a single value for each entity, such as a person's age.
- Multi-valued attribute: Attributes that can hold multiple values for each entity, such as a person's phone numbers.
- Derived attribute: Attributes whose values can be derived from other attributes within the database, rather than being stored explicitly.



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Q.3 Explain the role of join operation?

[2 marks]

Ans:- The need for the join operation in a relational database arises when data is distributed across multiple tables, and there's a requirement to retrieve related information from these tables simultaneously. Join operations allow users to combine rows from two or more tables based on a related column between them, enabling the extraction of meaningful information by correlating data from different sources. It's crucial for complex queries and data analysis tasks involving data stored across multiple tables.

Q.4 Illustrate all basic relational algebra operations. Consider following relations and analyze relational algebra Queries:

Suppliers (sID, sName, address)

Parts (pID, pName, colour)

Catalog (sID, pID, price)

(a) Find all prices for parts that are red or green. (A part may have different prices from different manufacturers.)

(b) Find the names of all suppliers who supply a part that is red or green

[4 marks]

Ans:- Relational algebra is a formal system for manipulating relations, which are analogous to tables in a relational database. The basic relational algebra operations include:

1. **Selection (σ):** This operation selects tuples from a relation that satisfy a given condition. It's denoted by $\sigma_{\text{condition}}(\text{Relation})$.
2. **Projection (π):** This operation selects certain columns (attributes) from a relation, discarding the others. It's denoted by $\pi_{\text{column}}(\text{Relation})$.
3. **Union ($\cup$):** This operation combines tuples from two relations, removing duplicates. It requires both relations to have the same set of attributes.
4. **Intersection ($\cap$):** This operation returns tuples that are common to both relations. Like union, it requires both relations to have the same set of attributes.
5. **Difference ($-$):** This operation returns tuples that are present in one relation but not in another. It also requires both relations to have the same set of attributes.
6. **Cartesian Product ($\times$):** This operation returns the combination of all possible pairs of tuples from two relations. It's denoted by $\text{Relation1} \times \text{Relation2}$.
7. **Join ($\bowtie$):** This operation combines tuples from two relations based on a common attribute. There are several types of joins such as Equi-join, Theta-join, Natural join, etc.

Now, let's use these operations to construct queries for your scenario:

Given the relations:



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- Suppliers (sID, sName, address)
- Parts (pID, pName, colour)
- Catalog (sID, pID, price)

(a) Find all prices for parts that are red or green:

To achieve this, we first need to join the Parts and Catalog relations based on the pID attribute, then select the tuples where the colour is red or green, and finally project the price attribute.

Query: $\pi \text{ price}(\sigma \text{ colour}='red' \vee \text{ colour}='green'(\text{Parts} \bowtie \text{Catalog}))$

This query joins the Parts and Catalog relations, filters the resulting tuples to only include those with a colour of red or green, and then projects the price attribute.

(b) Find the names of all suppliers who supply a part that is red or green:

To find the names of suppliers who supply red or green parts, we need to join the Suppliers, Parts, and Catalog relations based on common attributes and select the distinct supplier names.

Query: $\pi \text{ sName}(\sigma \text{ colour}='red' \vee \text{ colour}='green'(\text{Suppliers} \bowtie \text{Catalog} \bowtie \text{Parts}))$

This query joins the Suppliers, Catalog, and Parts relations, filters the resulting tuples to only include those with a part colour of red or green, and then projects the supplier names.

These queries demonstrate how to use basic relational algebra operations to retrieve specific information from a set of related tables.

Q.5 Demonstrate advantageous of DBMS over File System for any application.

[4 marks]

Ans:- Advantageous of DBMS over File System :-

1. Data Integrity: DBMS ensures data integrity through various constraints (such as primary key, foreign key, and check constraints), which prevent invalid data entry or corruption.
2. Data Security: DBMS provides robust security features like user authentication, authorization, and encryption to protect sensitive data from unauthorized access.
3. Concurrent Access: DBMS allows multiple users to access and modify data concurrently while maintaining data consistency, whereas file systems may face issues like data corruption or inconsistency with concurrent access.
4. Data Independence: DBMS offers both logical and physical data independence. Logical independence allows changes to the database schema without affecting the application, while physical independence allows changes to the physical storage structures without affecting the application.
5. Data Consistency and Atomicity: DBMS ensures that transactions are processed reliably with ACID properties (Atomicity, Consistency, Isolation, Durability), whereas in a file system, ensuring such properties would require custom implementation.
6. Query Language: DBMS provides a powerful query language (SQL) for retrieving and manipulating



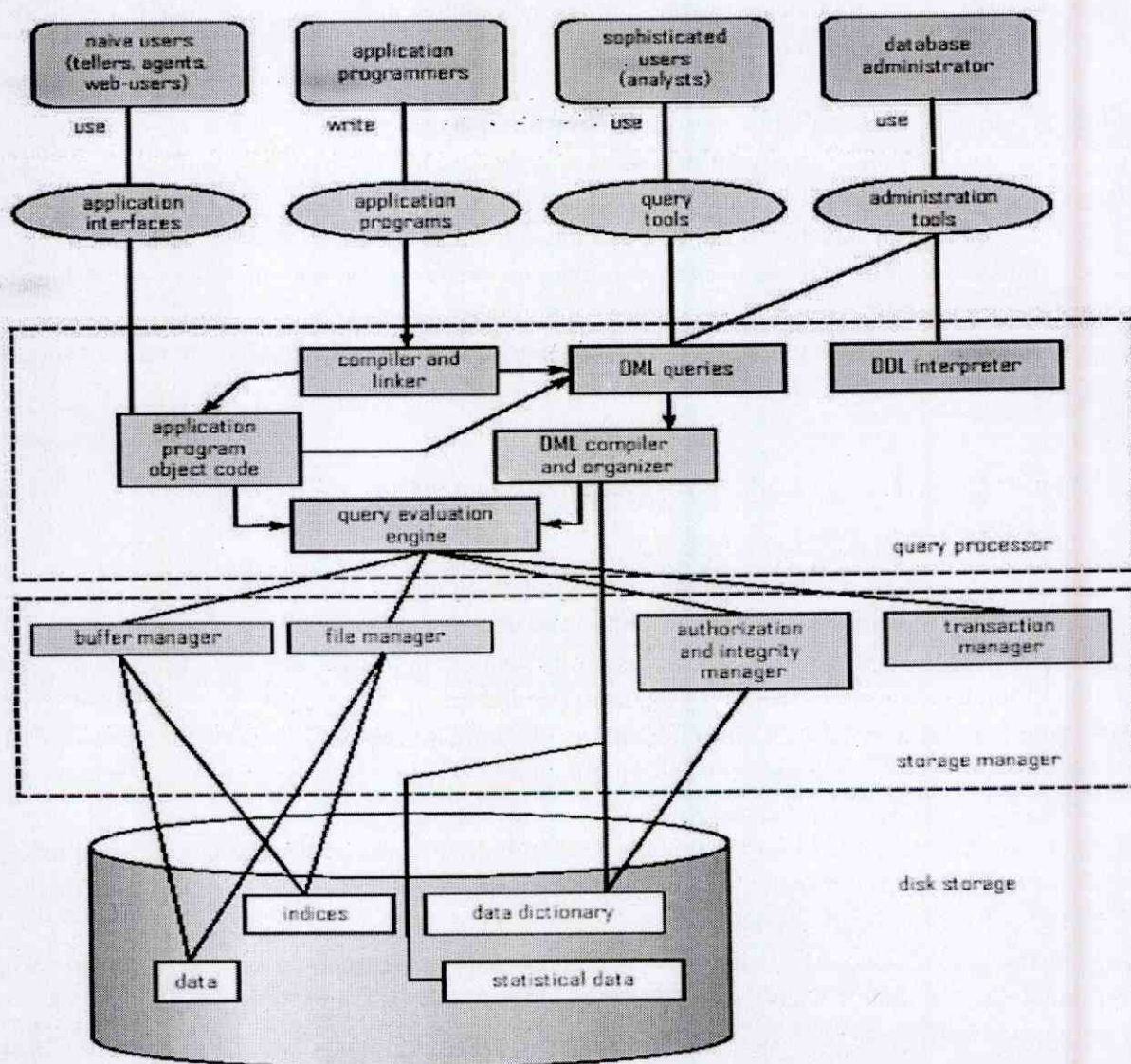
Solution of Question Paper
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data, making it easier to perform complex operations compared to file systems, where custom code would be needed for each operation.

7. Backup and Recovery: DBMS offers built-in features for backup and recovery, allowing administrators to restore the database to a consistent state after failures, which is more challenging with file systems.

Q.6. Show a neat Diagram of DBMS System Structure also outlines all the parts of it. [6 marks]





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1. Query Processor: It interprets the requests (queries) received from end user via an application program into instructions. It also executes the user request which is received from the DML compiler. Query Processor contains the following components –

- **DML Compiler:** It processes the DML statements into low level instruction (machine language), so that they can be executed.
- **DDL Interpreter:** It processes the DDL statements into a set of table containing meta data (data about data).
- **Embedded DML Pre-compiler:** It processes DML statements embedded in an application program into procedural calls.
- **Query Optimizer:** It executes the instruction generated by DML Compiler.

2. Storage Manager: Storage Manager is a program that provides an interface between the data stored in the database and the queries received. It is also known as Database Control System. It maintains the consistency and integrity of the database by applying the constraints and executing the DCL statements. It is responsible for updating, storing, deleting, and retrieving data in the database. It contains the following components –

- **Authorization Manager:** It ensures role-based access control, i.e., checks whether the particular person is privileged to perform the requested operation or not.
- **Integrity Manager:** It checks the integrity constraints when the database is modified.
- **Transaction Manager:** It controls concurrent access by performing the operations in a scheduled way that it receives the transaction. Thus, it ensures that the database remains in the consistent state before and after the execution of a transaction.
- **File Manager:** It manages the file space and the data structure used to represent information in the database.
- **Buffer Manager:** It is responsible for cache memory and the transfer of data between the secondary storage and main memory.

3. Disk Storage: It contains the following components –

- **Data Files:** It stores the data.
- **Data Dictionary:** It contains the information about the structure of any database object. It is the repository of information that governs the metadata.
- **Indices:** It provides faster retrieval of data item.

The structure of a Database Management System (DBMS) can be divided into three main components: the Internal Level, the Conceptual Level, and the External Level.

1. **Internal Level:** This level represents the physical storage of data in the database. It is responsible for storing and retrieving data from the storage devices, such as hard drives or solid-state drives. It deals with low-level implementation details such as data compression, indexing, and storage allocation.
2. **Conceptual Level:** This level represents the logical view of the database. It deals with the overall organization of data in the database and the relationships between them. It defines the data schema,



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Prog./Sem.: CSE, AI, DS, IoT, IT/ IV	Course Title: Database Management System	Course Code: 4CS4-05, 4IT4-05, 4CDS4-05, 4CAI4-05, 4CIT4-05
Duration: 1.5 hours	Date: 30/3/24 Session : FIRST/SECOND/THIRD	Max Marks: 20
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which includes tables, attributes, and their relationships. The conceptual level is independent of any specific DBMS and can be implemented using different DBMSs.

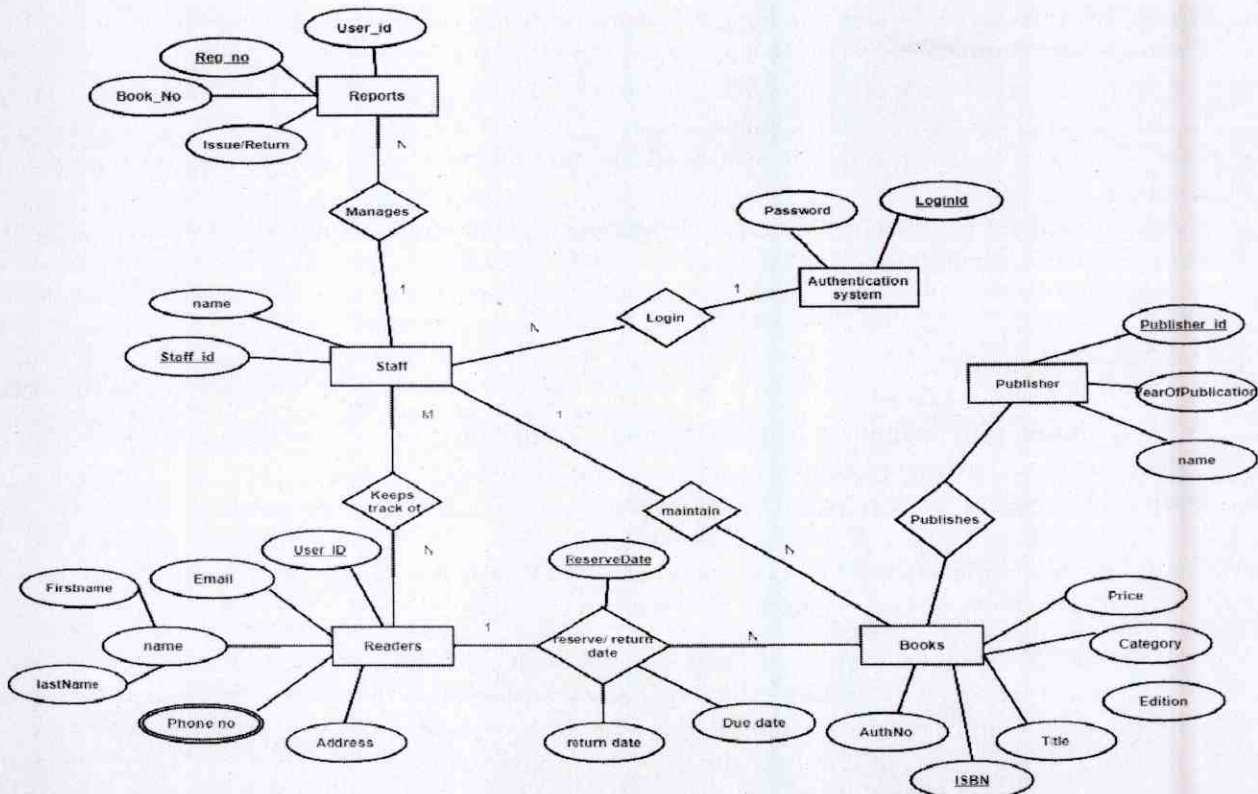
3. External Level: This level represents the user's view of the database. It deals with how users access the data in the database. It allows users to view data in a way that makes sense to them, without worrying about the underlying implementation details. The external level provides a set of views or interfaces to the database, which are tailored to meet the needs of specific user groups.

The three levels are connected through a schema mapping process that translates data from one level to another. The schema mapping process ensures that changes made at one level are reflected in the other levels.

In addition to these three levels, a DBMS also includes a Database Administrator (DBA) component, which is responsible for managing the database system. The DBA is responsible for tasks such as database design, security management, backup and recovery, and performance tuning.

Q.7 Model the E-R diagram for library management system with complete labelling. Convert the drawn E-R diagram into relational model. For each relation in relational model clearly specify primary key and foreign key. [6 marks]

Ans:-





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let's convert this E-R diagram into a relational model:-

Book:

ISBN (Primary Key)
Title
Author
Genre
Publisher
Publish Year
Available Copies

Author:

Author ID (Primary Key)
Name
Nationality
Member:
Member ID (Primary Key)
Name
Address
Contact Number

Librarian:

Librarian ID (Primary Key)
Name
Department

Publisher:

Publisher ID (Primary Key)
Name
Address

Genre:

Genre ID (Primary Key)
Name

Transaction:

Transaction ID (Primary Key)
Member ID (Foreign Key)
Librarian ID (Foreign Key)
Book ISBN (Foreign Key)
Date Borrowed
Date Returned

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I-Mid Term Examination, March-2024

Prog./Semester:	B. Tech.- IV	Branch:	CSE/AI/DS/IOT/IT
Course Title:	TOC	Course Code:	4CS4-06/4CAI4- 06/ 4AID4-06/ 4CIT4-06/ 4IT4-6
Duration:	1.5 Hours	Maximum Marks:	20
Session : First		Roll No.:	
Instructions if any:			

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	What do you mean by acceptance of string. Give example	2	1	1
2	Differentiate Moore and Mealy machines with mathematical representation.	2	1	2
3	Define DFA with mathematical representation.	2	1	2

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Apply the Arden's theorem and find the regular expression corresponding to the given F.A (q1 is initial state) ,	4	1	3

5	Consider a Mealy machine represented by given figure. Construct Moore machine equivalent to this Mealy machine.	4	2	3
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PART- C (Descriptive/Analytical/Problem solving/Design questions)
(Attempt any one Question)

Q. No	Question	Max. Marks	CO(s)	Bloom's Level																								
6	Construct the finite automata equivalent to the regular expression $(0+1)^*(00+11)(0+1)^*$	6	2	3																								
7	Construct a minimum state automation equivalent to a given automaton M as given in figure where q6 is a final state. <table><tr><th>States</th><th>Input a</th><th>Input b</th></tr><tr><td>$\rightarrow q_0$</td><td>q_0</td><td>q_3</td></tr><tr><td>q_1</td><td>q_2</td><td>q_5</td></tr><tr><td>q_2</td><td>q_3</td><td>q_4</td></tr><tr><td>q_3</td><td>q_0</td><td>q_5</td></tr><tr><td>q_4</td><td>q_0</td><td>q_6</td></tr><tr><td>q_5</td><td>q_1</td><td>q_4</td></tr><tr><td>q_6</td><td>q_1</td><td>q_3</td></tr></table>	States	Input a	Input b	$\rightarrow q_0$	q_0	q_3	q_1	q_2	q_5	q_2	q_3	q_4	q_3	q_0	q_5	q_4	q_0	q_6	q_5	q_1	q_4	q_6	q_1	q_3	6	2	4
States	Input a	Input b																										
$\rightarrow q_0$	q_0	q_3																										
q_1	q_2	q_5																										
q_2	q_3	q_4																										
q_3	q_0	q_5																										
q_4	q_0	q_6																										
q_5	q_1	q_4																										
q_6	q_1	q_3																										



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Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.: B.Tech(CSE/AI/DS/IOT/IT)/IV	Course Title: Theory of Computation	Course Code: 4CS4-06/4CAI4-06/4CDS4- 06/4CIT4-06/4IT4-06
Duration: 1.5 hours	Date: 1/4/2024 Session: FIRST	Max Marks: 20
Submitted By: Dr. Sunita Gupta, Dr. Mukesh Gupta, Mr. Ashish Pant, Mr. Pratipal Singh, Mr. Rajesh Rajaan, Dr. C.M. Choudhary, Mr. Manish Bhardwaj, Ms. Dolly Mittal.		

S. No.	Course Outcome	Percentage Coverage
C01	Define and discuss the concept of formal grammar, formal language, regular expression and automata machine.	38.46
C02	Design finite automata and push down automata (PDA) machines for given formal languages or computational real-world problem statements.	61.54
C03	Understand the capability of Turing machine & design Turing Machine for context-sensitive languages or computational real-world problem statements.	00
C04	Choose and design appropriate automata for modeling the solution for various computational engineering problems.	00
C05	Understand the concepts of tractable & intractable problems, and able to decide a given problem is tractable or not.	00

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	7.69
2	BL2- Understand	15.38
3	BL3- Apply	53.85
4	BL4- Analyze	23.08
5	BL5- Evaluate	00
6	BL6- Create	00

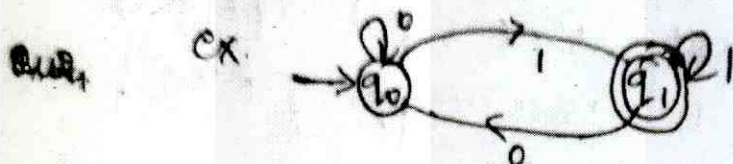
-- Question Paper Solution --



Solution of Question Paper
I Mid-Term Examination, March 2024

Branch/Semester: CSE/IT/AI/DS/IoT/IV	Subject: Theory of Computation	Subject Code: ACSA-06/ACAIA-06/ACDSA-06/ACITA-06/AITA-06
Duration: 1.5 hours	Date: 01/04/2024 Session (I/II/III): I	Max Marks: 20
Submitted By: <u>Dr. Sanita Gupta</u>		

Q.1. Accept of a string - A string x is accepted by a finite automata $(\theta, \Sigma, \delta, q_0, F)$, if $\delta(q_0, x) = q$ for some $q \in F$ (means acceptability by final state).



string. 01 is acceptable by this FA as -

$$\delta(q_0, 01) = \delta(q_0, 1) = (q_1, N).$$

Q.2. Moore m/c - is a six tuple $m/c(\theta, \Sigma, \Delta, \delta, \lambda, q_0)$

where 1) θ = finite set of states

Σ = set of input alphabets

Δ = set of output alphabets

δ = transition function $\Sigma \times \theta \rightarrow \theta$

λ = output function $\theta \rightarrow \Delta$

q_0 = initial state.

Mealy m/c - Mealy m/c is a six tuple $m/c(\theta, \Sigma, \Delta, \delta, \lambda, q_0)$ where all symbols have the same meaning except λ .

λ is output function mapping $\Sigma \times \theta \rightarrow \Delta$.



Solution of Question Paper
I Mid-Term Examination, March -2024

Branch/Semester: CSE/IT/AI/DS/IoT/IV	Subject: Theory of Computation	Subject Code-4CS4-06/4CAI4-06/4CDS4-06/4CIT4-06/4IT4-06
Duration: 1.5 hours	Date: 01/04/2024 Session (I/II/III): I	Max Marks: 20
Submitted By: <i>Dr. Sunita Gupta</i>		

Q.3 DFA - A finite automata is defined using 5 tuples $(Q, \Sigma, \delta, q_0, F)$, where

- (a) Q is finite nonempty set of states
- (b) Σ is finite nonempty set of input alphabets
- (c) δ is a transition function mapping $Q \times \Sigma \rightarrow Q$
- (d) $q_0 \in Q$ is the initial state
- (e) $F \subseteq Q$ is the set of final states

Q.4 $q_1 = q_{10} + q_{40} + q_{30} + \lambda \quad \text{--- (1)}$

$q_2 = q_{21} + q_{11} + q_{41} \quad \text{--- (2)}$

$q_3 = q_{20} \quad \text{--- (3)}$

$q_4 = q_{31} \quad \text{--- (4)}$

$q_4 = q_{31} = q_{201}$

Now

$q_2 = q_{21} + q_{11} + q_{2011} = q_{21}(1 + 011) + q_{11}$

using Arden's theorem,

$q_2 = q_{11}(1 + 011)^*$

Now $q_1 = q_{10} + q_{2010} + q_{200} + \lambda$

$= q_{10} + q_{21}(010 + 00) + \lambda = q_{10} + q_{11}(1 + 011)(010 + 00) + \lambda$

$q_1 = q_{11}(0 + 1(1 + 011)^*(00 + 010)) + \lambda$



Solution of Question Paper
I Mid-Term Examination, March -2024

Branch/Semester: CSE/IT/AI/DS/IoT/IV	Subject: Theory of Computation	Subject Code-4CS4-06/4CAI4-06/4CDS4-06/4CIT4-06/4IT4-06
Duration: 1.5 hours	Date: 01/04/2024 Session (I/II/III): I	Max Marks: 20
Submitted By: <i>Dr. Sanita Gupta</i>		

using Arden's theorem

$$q_1 = \wedge (0+1(1+011)^* (00+010))^*$$

$$\text{Now } q_4 = q_1 01 = q_1 (1+011)^* 01$$

$$q_4 = (0+1(1+011)^* (00+010))^* (1+011)^* 01$$

So Regular expression corresponding to the given FA is -

$$(0+1(1+011)^* (00+010))^* (1+011)^* 01$$

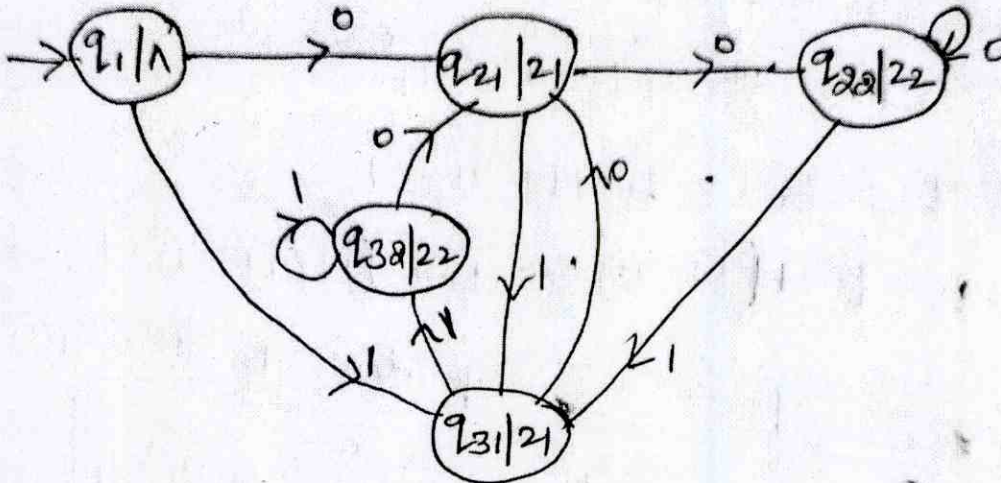
Q.5 Moore m/c corresponding to the given mealy m/c is -

PS	NS		O/P
	q=0	q=1	
→ q ₁	q ₂₁	q ₃₁	∧
q ₂₁	q ₂₂	q ₃₁	21
q ₂₂	q ₂₂	q ₃₁	22
q ₃₁	q ₂₁	q ₃₂	21
q ₃₂	q ₂₁	q ₃₂	22

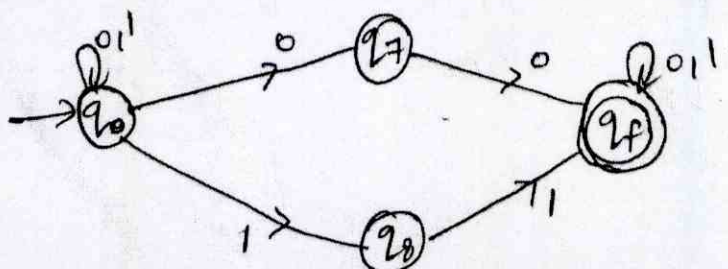
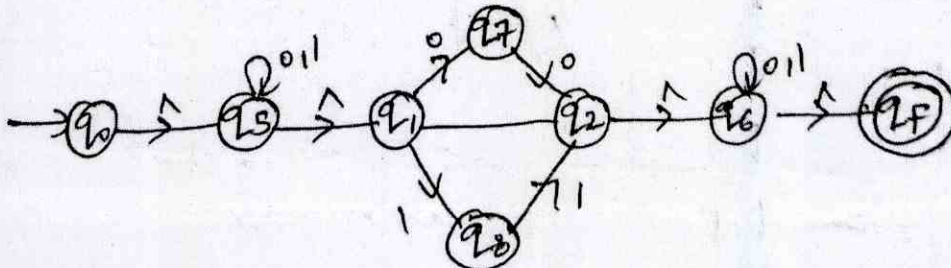
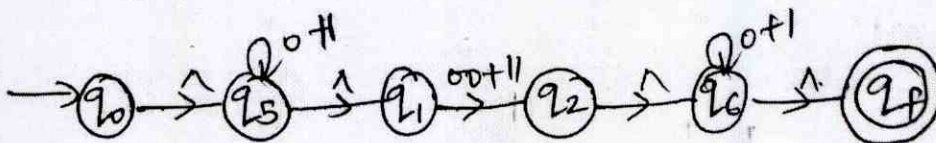
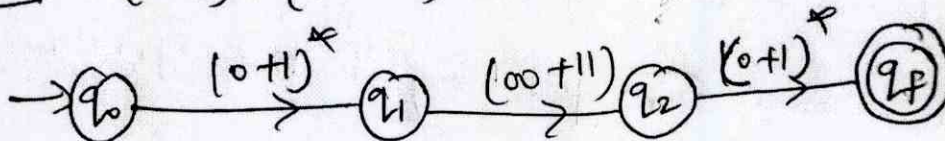


Solution of Question Paper
I Mid-Term Examination, March -2024

Branch/Semester: CSE/IT/AI/DS/IoT/IV	Subject: Theory of Computation	SubjectCode-4CS4-06/4CAI4-06/4CD54-06/4CIT4-06/4IT4-06
Duration: 1.5 hours	Date:01/04/2024 Session (I/II/III): I	Max Marks: 20
Submitted By: <i>Dr. Savita Gupta</i>		



Q.6. $(0+1)^* (00+11) (0+1)^*$





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Branch/Semester: CSE/IT/AI/DS/IoT/IV	Subject: Theory of Computation	SubjectCode-4CS4-06/4CAI4-06/4CDS4-06/4CIT4-06/4IT4-06
Duration: 1.5 hours	Date:01/04/2024 Session (I/II/III): I	Max Marks: 20
Submitted By: Dr. Sanita Gupta		

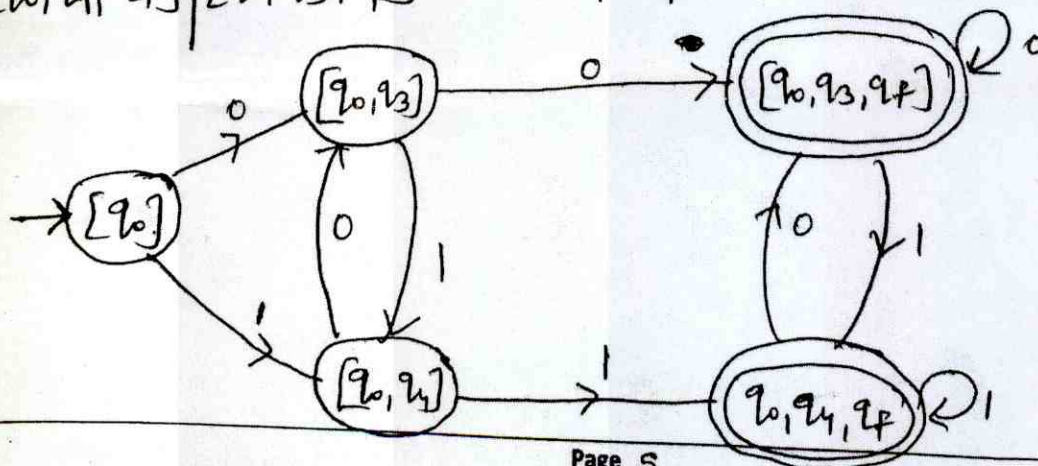
Now Convert it into DFA as it is NDFA.

Table of NDFA is -

State	0	1
→q ₀	q ₀ , q ₃	q ₀ , q ₄
q ₃	q _f	
q ₄	-	q _f
q _f	q _f	q _f

Transition table of DFA is -

state	0	1
→[q ₀]	[q ₀ , q ₃]	[q ₀ , q ₄]
[q ₀ , q ₃]	[q ₀ , q ₃ , q _f]	[q ₀ , q ₄]
[q ₀ , q ₄]	[q ₀ , q ₃]	[q ₀ , q ₄ , q _f]
[q ₀ , q ₃ , q _f]	[q ₀ , q ₃ , q _f]	[q ₀ , q ₄ , q _f]
[q ₀ , q ₄ , q _f]	[q ₀ , q ₃ , q _f]	[q ₀ , q ₄ , q _f]





Solution of Question Paper
I Mid-Term Examination, March -2024

Branch/Semester: CSE/IT/AI/DS/IoT/IV	Subject: Theory of Computation	Subject Code-ACSA-06/ACAIA-06/ACD54-06/ACITA-06/AITA-06
Duration: 1.5 hours	Date:01/04/2024 Session (I/II/III): I	Max Marks: 20
Submitted By: <i>Ar. Sumita Gupta</i>		

Q.7 $\pi_0 = \{ \{q_6\}, \{q_0, q_1, q_2, q_3, q_4, q_5\} \}$

$\pi_1 = \{ \{q_6\}, \{q_0, q_1, q_2, q_3, q_5\}, \{q_4\} \}$

$\pi_2 = \{ \{q_6\}, \{q_4\}, \{q_0, q_1, q_3\}, \{q_2, q_5\} \}$

$\pi_3 = \{ \{q_6\}, \{q_4\}, \{q_0\}, \{q_1\}, \{q_3\}, \{q_2, q_5\} \}$

$\pi_4 = \{ \{q_6\}, \{q_4\}, \{q_0\}, \{q_1\}, \{q_3\}, \{q_2\}, \{q_5\} \}$

Here $\pi = \emptyset$. The minimum state automaton is simply the given automaton.



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I-Mid Term Examination, March-2024

Prog./Semester:	B. Tech.- IV	Branch:	CSE/CSE(AI)/CSE(DS)/CSE(IOT)/IT
Course Title:	DCCN	Course Code:	4CS/CAI/CDS/CIT4-07/4IT4-07
Duration:	1.5 Hours	Maximum Marks:	20
Session : Second	Roll No.:		

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	Explain each modulation techniques with the help of diagram	2	1	2
2	Describe Stop & Wait ARQ protocol.	2	2	2
3	Explain pure and slotted ALOHA.	2	2	2

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Illustrate OSI model. Explain any two functions of each layer in the OSI model.	4	1	3
5	A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is $x^4 + x + 1$. Solve for actual transmitted bit.	4	2	3

PART -C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	Design the waveform for each of the following codes for the bit stream 01001110 a)NRZ-I b)Manchester c)Differential Manchester d) AMI e) Unipolar f) NRZ	6	1	5
7	Analyze the various transmission medium used for transmitting any data on computer network.	6	1	4



Solution of Question Paper
I Mid-Term Examination, March -2024

Prog./Sem.: IV	Course Title: B-Tech DCCN	CourseCode: 4CS4/4CAI4/4CDS4/4CIT4- 07/4IT4-07
Duration: 1.5 hours	Date: 01/04/2024 Session : SECOND	Max Marks: 20
Submitted By: Dr Megha Gupta, M.K.Beniwal, Priya Gupta, Alisha Goyal, Nimish Arvind, Nikhar Bhatnagar, Shikha Srivastava		

S. No.	Course Outcome	Percentage Coverage
CO1	Understand the concept of Signals, OSI & TCP/IP reference models and functionalities of each layer in these models.	69.23%
CO2	Apply flow control and error control mechanisms in data link layer protocols.	30.76%
CO3	Apply the concepts of various routing algorithms and IP addresses to fulfil network requirements of an organization.	NA
CO4	Analyse the model of reliable/unreliable communication using various Transport Layer Protocols (UDP, TCP).	NA
CO5	Analyse the features and operations of various application layer protocols such as HTTP, DNS and SMTP.	NA

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	NA
2	BL2- Understand	23.08%
3	BL3- Apply	55.84
4	BL4- Analyze	23.07%
5	BL5- Evaluate	NA
6	BL6- Create	NA

Question Paper Solution --

Q.1 Explain each modulation techniques with the help of diagram?

Ans:

Amplitude Modulation

In this modulation, the amplitude of the carrier signal is modified to reflect the analog data. Amplitude modulation is implemented by means of a multiplier. The amplitude of modulating signal (analog data) is multiplied by the amplitude of carrier frequency, which then reflects analog data. The frequency and phase of carrier signal remain unchanged.

Frequency Modulation



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**Solution of Question Paper
I Mid-Term Examination, March -2024**

Branch/Semester: CSE, AI, DS, IT, IOT	Subject: DCCN	Subject Code: 4CS/CAI/CDS/CIT4- 07/4IT4-07
Duration: 1.5 hours	Date: 01.04.2024 Session : II	Max Marks: 20
Submitted By: Megha Gupta, Priya Gupta, Mahender, Nikhar Bhatnagar, Nimish Arvind, Sikha		

Q.1 Explain each modulation techniques with the help of diagram?

Ans:

Amplitude Modulation

In this modulation, the amplitude of the carrier signal is modified to reflect the analog data. Amplitude modulation is implemented by means of a multiplier. The amplitude of modulating signal (analog data) is multiplied by the amplitude of carrier frequency, which then reflects analog data. The frequency and phase of carrier signal remain unchanged.

Frequency Modulation

In this modulation technique, the frequency of the carrier signal is modified to reflect the change in the voltage levels of the modulating signal (analog data).

Phase Modulation

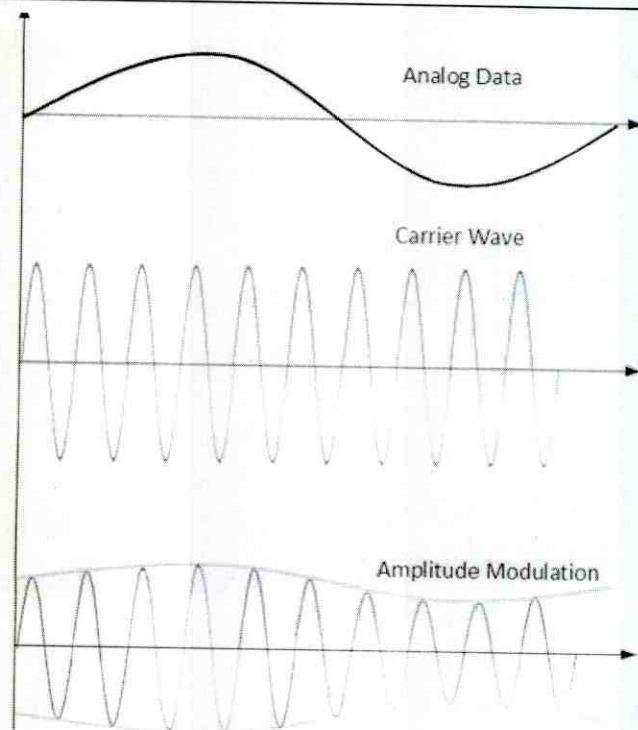
In the modulation technique, the phase of carrier signal is modulated in order to reflect the change in voltage (amplitude) of analog data signal. Phase modulation is practically similar to Frequency Modulation, but in Phase modulation frequency of the carrier signal is not increased. Frequency of carrier is signal is changed (made dense and sparse) to reflect voltage change in the amplitude of modulating signal.



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**Solution of Question Paper
I Mid-Term Examination, March -2024**

Branch/Semester: CSE, AI, DS, IT, IOT	Subject: DCCN	Subject Code: 4CS/CAI/CDS/CIT4- 07/4IT4-07
Duration: 1.5 hours	Date: 01.04.2024 Session : II	Max Marks: 20
Submitted By: Megha Gupta, Priya Gupta, Mahender, Nikhar Bhatnagar, Nimish Arvind, Sikha		



Q.2 Describe Stop & Wait ARQ protocol.

Ans:

Stop-and-Wait Automatic Repeat reQuest (ARQ) is a simple protocol used for reliable data transmission over a communication channel that may introduce errors or loss. It ensures that each packet of data sent by the sender is received and acknowledged by the receiver before the next packet is sent. Here's a description of the Stop-and-Wait ARQ protocol:

Sender-Side Operation:

The sender divides the data into fixed-size packets before transmission. After sending each packet, the sender waits for an acknowledgment (ACK) from the receiver. If the sender receives an ACK within a specified timeout period, it proceeds to send the next packet. If the sender does not receive an ACK within the timeout period, it assumes that the packet was lost or corrupted and retransmits the same packet.

Receiver-Side Operation:



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Duration: 1.5 hours	Date: 01.04.2024 Session : II	Max Marks: 20
Submitted By: Megha Gupta, Priya Gupta, Mahender, Nikhar Bhatnagar, Nimish Arvind, Sikha		

The receiver waits for a packet to arrive from the sender. Upon receiving a packet, the receiver checks for errors and verifies the sequence number to ensure correct ordering. If the received packet is error-free and in the correct sequence, the receiver sends an ACK back to the sender. If the packet is corrupted or out of order, the receiver discards the packet and does not send an ACK. This triggers the sender to retransmit the packet.

Timeout Handling:

Both the sender and receiver use timeouts to detect lost or delayed packets. If the sender's timeout expires before receiving an ACK, it assumes that the packet was lost and retransmits it. If the receiver's timeout expires without receiving a packet, it sends a negative acknowledgment (NAK) or remains silent, depending on the protocol variant.

Flow Control:

Stop-and-Wait ARQ provides a form of flow control by ensuring that the sender does not overwhelm the receiver with too many packets. The sender waits for an ACK before sending the next packet, ensuring that the receiver has sufficient time to process and acknowledge each packet.

Handling Duplicate Packets:

The receiver may encounter duplicate packets due to retransmissions by the sender. To avoid processing duplicate packets, the receiver checks the sequence number of each received packet and discards duplicates.

Performance Considerations:

Stop-and-Wait ARQ is simple but inefficient for high-latency or high-bandwidth networks due to its low throughput. It introduces additional delays as the sender must wait for acknowledgments before sending the next packet. For better performance, variants like Selective Repeat ARQ or Go-Back-N ARQ are used, which allow the sender to transmit multiple packets before receiving acknowledgments.

Overall, Stop-and-Wait ARQ provides a basic mechanism for reliable data transmission in scenarios where simplicity and reliability are prioritised over throughput efficiency.



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**Solution of Question Paper
I Mid-Term Examination, March -2024**

Branch/Semester: CSE,AI,DS,IT,IOT	Subject: DCCN	Subject Code: 4CS/CAI/CDS/CIT4- 07/4IT4-07
Duration: 1.5 hours	Date: 01.04.2024 Session : II	Max Marks: 20
Submitted By: Megha Gupta, Priya Gupta, Mahender, Nikhar Bhatnagar, Nimish Arvind, Sikha		

Q.3 Differentiate between pure and slotted ALOHA.

Ans: Pure ALOHA and Slotted ALOHA are both random access protocols used in packet-switched networks, but they differ in how they handle the timing of packet transmissions and collision resolution. Here's a differentiation between the two:

Timing:

Pure ALOHA: In Pure ALOHA, each station can transmit a packet at any time, without synchronisation with other stations or a predefined time slot.

Slotted ALOHA: Slotted ALOHA divides time into discrete slots, and stations are only allowed to transmit at the beginning of each time slot. The timing is synchronised among all stations.

Transmission Window:

Pure ALOHA: Stations can initiate transmission at any time, leading to potential collisions between packets. There is no coordination or restriction on when stations can transmit.

Slotted ALOHA: Stations can only transmit at the start of each time slot. This reduces the likelihood of collisions as transmissions are synchronised.

Collision Resolution:

Pure ALOHA: Collisions occur when two or more stations transmit packets simultaneously. When a collision is detected, the stations involved wait for a random amount of time before retransmitting their packets.

Slotted ALOHA: Collisions can still occur, but they are confined within specific time slots. Stations can detect collisions by monitoring the channel during their assigned time slots. If a collision is detected, stations wait until the next time slot to retransmit their packets.

Efficiency:

Pure ALOHA: Pure ALOHA has lower efficiency compared to Slotted ALOHA due to the higher probability of collisions. The maximum achievable throughput is around 18.4%.



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Branch/Semester: CSE,AI,DS,IT,IOT	Subject: DCCN	Subject Code: 4CS/CAI/CDS/CIT4- 07/4IT4-07
Duration: 1.5 hours	Date: 01.04.2024 Session : II	Max Marks: 20
Submitted By: Megha Gupta, Priya Gupta, Mahender, Nikhar Bhatnagar, Nimish Arvind, Sikha		

Slotted ALOHA: Slotted ALOHA is more efficient than Pure ALOHA as it reduces the probability of collisions by synchronising transmissions. The maximum achievable throughput is around 37%.

Implementation Complexity:

Pure ALOHA: Pure ALOHA is simpler to implement as it does not require time slot synchronisation.

Slotted ALOHA: Slotted ALOHA requires synchronisation among stations to ensure that transmissions occur at the beginning of each time slot, which adds some complexity to the implementation.

Overall, Slotted ALOHA offers higher efficiency and reduced collision probability compared to Pure ALOHA due to its synchronised time slots. However, it requires additional overhead for slot synchronisation, which can affect its simplicity and implementation complexity.

Q.4 Illustrate OSI model. Explain any two functions of each layer in the OSI model.

Ans:

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes the functions of a telecommunication or computing system into seven distinct layers. Each layer serves a specific purpose and provides services to the layers above and below it. Here's an illustration of the OSI model along with explanations of two functions of each layer:

Physical Layer (Layer 1):

Functions:

Physical Connection: The physical layer is responsible for establishing and terminating physical connections between devices. It defines the characteristics of the transmission medium, such as voltage levels, cable types, and data rates.

Data Encoding and Transmission: It converts digital data into signals suitable for transmission over the physical medium. This includes techniques like modulation, encoding, and line coding to ensure reliable data transfer.

Data Link Layer (Layer 2):



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Duration: 1.5 hours	Date: 01.04.2024 Session : II	Max Marks: 20
Submitted By: Megha Gupta, Priya Gupta, Mahender, Nikhar Bhatnagar, Nimish Arvind, Sikha		

Functions:

Frame Synchronization: The data link layer divides data into frames and provides mechanisms for synchronising the sender and receiver's frame boundaries. It adds headers and trailers to frames for error detection and correction.

Medium Access Control (MAC): This layer manages access to the physical medium and resolves conflicts when multiple devices attempt to transmit simultaneously. MAC protocols, such as CSMA/CD (Carrier Sense Multiple Access with Collision Detection), control the transmission of data frames.

Network Layer (Layer 3):

Functions:

Routing: The network layer determines the best path for data packets to travel from the source to the destination across multiple networks. It uses routing algorithms and maintains routing tables to make forwarding decisions.

Logical Addressing: Each device on a network is assigned a logical address (e.g., IP address) by the network layer. These addresses are used to uniquely identify devices and route data packets to their destinations.

Transport Layer (Layer 4):

Functions:

Segmentation and Reassembly: The transport layer breaks down large data streams into smaller segments for efficient transmission over the network. It also reassembles these segments at the receiving end to reconstruct the original data.

Error Detection and Correction: This layer provides mechanisms for error detection and correction to ensure reliable data delivery. It uses techniques like checksums and sequence numbers to detect and recover from transmission errors.

Session Layer (Layer 5):

Functions:



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Session Establishment and Termination: The session layer establishes, maintains, and terminates communication sessions between applications. It handles session setup, coordination, and synchronization, allowing applications to exchange data in an organised manner.

Dialog Control: This layer manages the dialogue between communicating devices, ensuring that data exchange occurs in an orderly fashion. It controls the flow of data, handles checkpoints, and supports error recovery mechanisms.

Presentation Layer (Layer 6):

Functions:

Data Translation: The presentation layer is responsible for translating data between different formats and representations to ensure interoperability between systems. It handles tasks like data encryption, compression, and data format conversions.

Data Compression: It compresses data to reduce bandwidth usage and improve transmission efficiency. Compression techniques such as Huffman coding and Lempel-Ziv-Welch (LZW) compression are used to reduce the size of data before transmission.

Application Layer (Layer 7):

Functions:

Application Services: The application layer provides interfaces and services for application-level communication and interaction. It includes protocols and standards that enable specific applications, such as HTTP for web browsing, SMTP for email, and FTP for file transfer.

User Interface: This layer presents data to the user and allows users to interact with applications. It provides user interfaces for accessing network resources, managing files, and performing other application-specific tasks.

Q.5 A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is $x^4 + x + 1$. Solve for actual transmitted bit.

Ans: To find the actual bit string transmitted using the CRC method, we need to perform CRC division. Here are the major steps:



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Solution of Question Paper
I Mid-Term Examination, March -2024

Branch/Semester: CSE,AI,DS,IT,IOT	Subject: DCCN	Subject Code: 4CS/CAI/CDS/CIT4- 07/4IT4-07
Duration: 1.5 hours	Date: 01.04.2024 Session : II	Max Marks: 20
Submitted By: Megha Gupta, Priya Gupta, Mahender, Nikhar Bhatnagar, Nimish Arvind, Sikha		

1. Append four zeros (degree of the polynomial generator) to the bit stream to create the augmented bit stream: 11010110110000

Divide the polynomial generator, $x^4 + x + 1$, into the augmented bit stream using modulo-2 division. This involves performing XOR operations at each step.

```
1. 11010110110000
2. -----
3.  $x^4 + x + 1$ ) 11010110110000
4.   1001
5.   ----
6.   101101
7.   1001
8.   ----
9.   11101
10.      1001
11.      ----
12.      1010
```

1. The remainder of the division is the CRC, which is 1010.
 2. Append the CRC to the original bit stream to get the transmitted bit string: 11010110111010.
- Therefore, the actual bit string transmitted is 11010110111010.

Q.6 Design the waveform for each of the following codes for the bit stream 01001110 a)NRZ-I b)Manchester c)Differential Manchester d) AMI e) Unipolar f) NRZ.

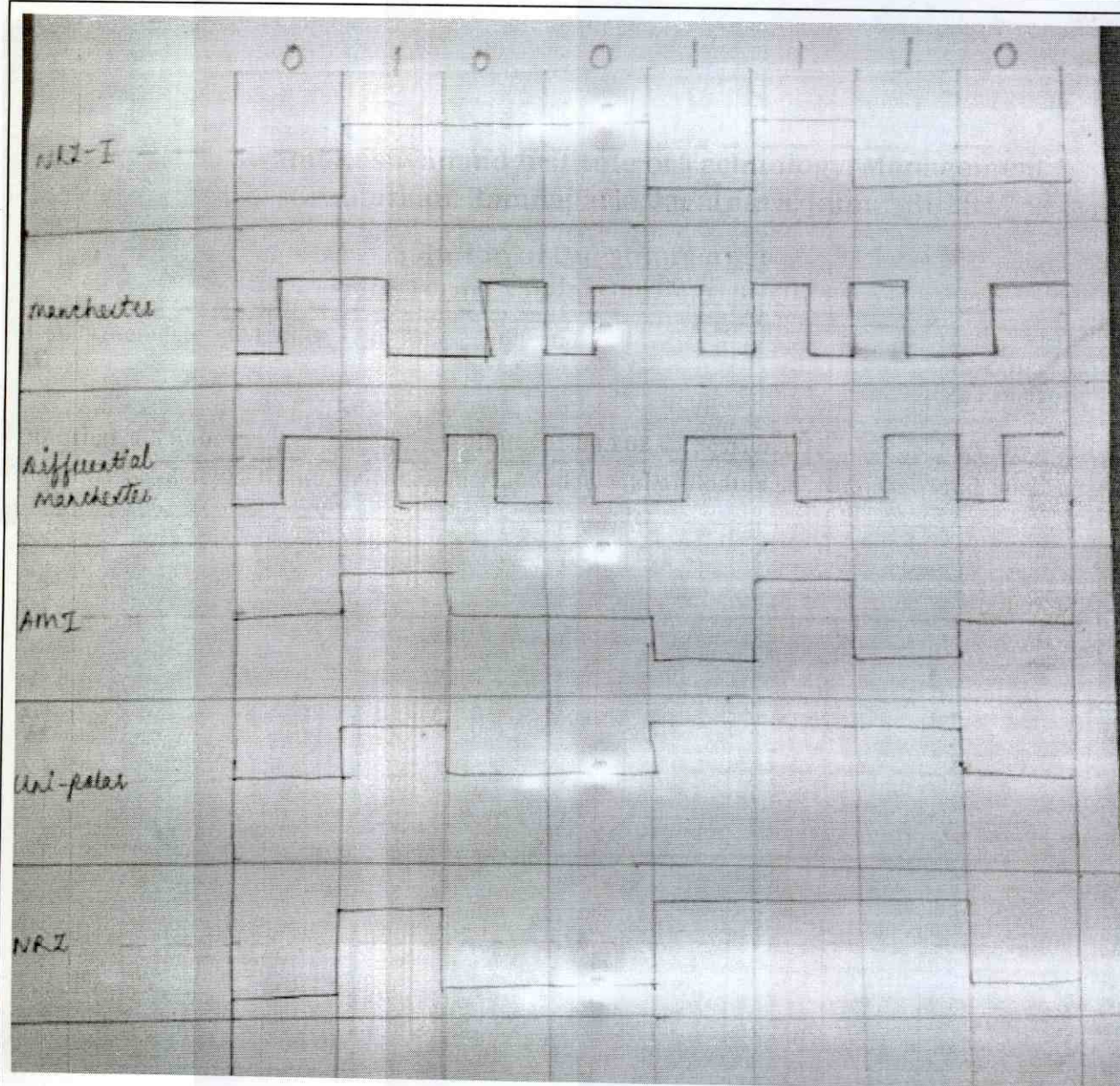
Ans:



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Q.7 Analyze the various transmission medium used for transmitting any data on computer network.

Ans: Various transmission media are used for transmitting data in computer networks. Each transmission medium has its characteristics, advantages, and limitations. Here's an analysis of some commonly used transmission media:

Twisted Pair Cable:

Type: Twisted pair cables consist of pairs of insulated copper wires twisted together.

Characteristics: They are flexible, inexpensive, and easy to install. Twisting helps reduce electromagnetic interference (EMI) and crosstalk.

Advantages: Widely used in telephone lines and Ethernet networks. Available in two types: unshielded twisted pair (UTP) and shielded twisted pair (STP).

Limitations: Susceptible to signal degradation over long distances and susceptible to EMI in environments with high electromagnetic interference.

Coaxial Cable:

Type: Coaxial cables consist of a central copper conductor surrounded by an insulating layer, a metallic shield, and an outer insulating layer.

Characteristics: Provides better shielding against EMI compared to twisted pair cables. Supports higher bandwidths and longer distances.

Advantages: Suitable for high-speed data transmission, such as cable television (CATV) networks and broadband internet.

Limitations: More expensive and less flexible than twisted pair cables. Susceptible to signal loss over long distances.

Fiber Optic Cable:



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Type: Fiber optic cables use optical fibers made of glass or plastic to transmit data as pulses of light.

Characteristics: Immune to electromagnetic interference. Capable of transmitting data over long distances at very high speeds.

Advantages: Provides high bandwidth, low attenuation, and secure transmission. Ideal for long-distance communications and high-speed networks.

Limitations: Expensive to install and maintain. Requires specialized equipment for installation and repair.

Wireless Transmission:

Type: Wireless transmission uses electromagnetic waves to transmit data without physical cables.

Characteristics: Offers mobility and flexibility. Includes technologies such as Wi-Fi, Bluetooth, and cellular networks.

Advantages: Eliminates the need for physical cables, suitable for mobile devices and remote locations. Offers scalability and easy deployment.

Limitations: Susceptible to interference and signal attenuation due to obstacles and distance. Limited bandwidth compared to wired connections.

Satellite Communication:

Type: Satellite communication involves transmitting data via communication satellites orbiting the Earth.

Characteristics: Enables long-distance communication across vast areas, including remote regions.

Advantages: Provides global coverage, suitable for communication in remote or inaccessible areas. Supports broadcast and multicast services.

Limitations: Higher latency compared to terrestrial transmission. Vulnerable to environmental factors such as weather conditions.